

Gemini 3.1 Flash Text-to-Speech Studio

— Complete Operator Manual

Welcome to the official User & Technical Manual for **Gemini 3.1 Flash TTS Studio**. This manual covers operational workflows, performance directing techniques, and underlying engineering architecture—structured progressively from **Beginner** to **Expert**.

Executive Overview & Core Capabilities

Gemini 3.1 Flash TTS Studio is a web application designed to synthesize natural, low-latency, and emotionally expressive speech powered by the gemini-3.1-flash-tts-preview API model.

Key Features Matrix

Feature	Description	Level
30 Prebuilt Voices	12 Female, 15 Male, and 3 Versatile voices categorized by pitch, timbre, and use-case.	Beginner
Inline Expressive Audio Tags	Micro-directing performance using tags like [whispers], [laughs], [gasp], and [excited].	Intermediate
Director Prompt Stage	High-level scene, persona, and environmental context injection.	Advanced
Smart Sentence Windows	Automatic 2,000-character sentence boundary chunking with pre-buffering.	Expert
Multi-Speaker Studio	Turn-based script builder for two distinct speakers utilizing single-pass API calls.	Intermediate
PCM16-to-WAV Engine	Client-side 44-byte RIFF header creation around raw	Expert

	signed 16-bit PCM streaming audio.	
--	---------------------------------------	--

1. Quick-Start Guide (Beginner)

Navigating the Interface

The interface consists of four primary workspace tabs and a persistent bottom Audio Dock:

- 1. **Solo Studio:** The main workspace for single-voice scripts, prompt directing, and inline tag insertion.
- 2. **Dialogue Builder:** Create multi-person audio scripts and podcasts.
- 3. **Voice Matrix:** Search, filter, and audition all 30 prebuilt Gemini neural voices.
- 4. **Library:** View, replay, and download WAV files generated during your current session.

Step-by-Step: Your First Synthesis

+-----+ +-----+ +-----+

| 1. Select a Voice | ---> | 2. Enter Text Script | ---> | 3. Click Synthesize |

| (e.g., Kore / Charon) | | (or click Quick Demo) | | (Audio plays in Dock) |

+-----+ +-----+ +-----+

- 1. Navigate to **Solo Studio**.
- 2. Select your desired speaker from the **Active Speaker Voice** dropdown (e.g., Kore for authoritative news, or Aoede for natural storytelling).
- 3. Type or paste your text into the script box, or click the **"3.1 Expressive Tags Demo"** preset button under *Expressive Samples*.
- 4. Click **Synthesize 3.1 Flash Speech**.
- 5. Your audio will render into a WAV stream and immediately begin playing in the bottom **Audio Player Dock**.

Audio Dock & Export Controls

The persistent bottom player provides total control over playback:

- **Play / Pause:** Click the center circle button or press Spacebar.
- **Playback Speed:** Click 1.0x to cycle through 0.75x, 1.0x, 1.25x, 1.5x, and 2.0x.
- **Waveform Visualizer:** Real-time canvas oscilloscope visualizing frequency output.
- **Export WAV:** Downloads the currently loaded track as a uncompressed .wav file.

2. Voice Matrix & Catalog Reference (Beginner–Intermediate)

The studio includes **30 prebuilt neural voices**. Choose voices based on vocal timbre, speed,

and target medium:

Female Voices (12)

Voice Name	Vocal Characteristics	Optimal Medium / Style
Kore	Firm, authoritative, structured	News, corporate trailers, public address
Aoede	Breezy, warm, conversational	Vlogs, podcasts, casual storytelling
Callirrhoe	Easy-going, relaxed, melodic	Audiobooks, guided relaxation, long-form reading
Autonoe	Bright, upbeat, energetic	Commercials, promotional ads, fitness apps
Leda	Youthful, dynamic, expressive	E-learning modules, animation, children's books
Algieba	Smooth, polished, professional	Product demonstrations, executive keynotes
Despina	Warm, inviting, clear	Virtual assistants, IVR phone systems
Erinome	Articulate, precise, balanced	Explainer videos, museum audio guides
Vindemiatrix	Soft, gentle, soothing	Bedtime stories, sleep audio, meditation
Sadachbia	Expressive, lively, engaging	Interactive fiction, gaming NPCs
Sulafat	Intimate, warm, narrative	Personal notes, literary narrations
Pulcherrima	Crisp, forward, direct	Public service announcements, alerts

Male Voices (15)

Voice Name	Vocal Characteristics	Optimal Medium / Style
Charon	Deep, resonant, informative	Documentaries, video essays, academic content
Fenrir	Intense, excitable, high-energy	Gaming trailers, sports broadcasts, action scenes
Orus	Firm, resonant, commanding	Movie trailers, official declarations
Enceladus	Breathy, intimate, dramatic	Character monologues, theatrical readings
Iapetus	Clear, standard, neutral	Technical tutorials, instructional guides
Umbriel	Casual, friendly, accessible	Podcast co-host, friendly explainer videos
Algenib	Gravelly, gritty, mature	Vintage narration, hardboiled fiction
Rasalgethi	Precise, analytical, structured	Tech briefings, news roundups
Achernar	Soft, calm, gentle	Poetry, reflective prose, ambient soundscapes
Alnilam	Professional, solid, calm	Financial analysis, corporate reports
Schedar	Even, flat, clean	Screen reader style, dictionary definitions
Gacrux	Wise, mature, warm	Historical documentaries, legend recitals
Achird	Approachable, bright, casual	Daily morning briefings, conversational AI

Zubenelgenubi	Unscripted, relaxed, slow	Casual dialogue, fireside chats
Sadaltager	Engaging, articulate, clear	University lectures, educational podcasts

Versatile / Neutral Voices (3)

Voice Name	Vocal Characteristics	Optimal Medium / Style
Zephyr	Bright, flexible, adaptive	Assistive technologies, system pop-ups
Puck	Playful, mischievous, dynamic	Creative projects, cartoon characters
Aoede-Lite	Breezy, light, clear	General-purpose voice assistant

3. Gemini 3.1 Expressive Audio Tags (Intermediate)

Gemini 3.1 Flash TTS allows **inline performance tags** inside your script. The TTS engine interprets these tags as expressive performance modifications rather than reading them aloud.

Available Audio Tags & Quick Insertion

Click any tag button above the text area in Solo Studio to insert it at your cursor position:

Tag Chip	Delivery Effect	Best Practice Code Example
[whispers]	Lowers volume, increases breathiness	[whispers] Keep quiet, they are right outside.
[laughs]	Injects natural laughter/chuckle	[laughs] I can't believe you actually tried that!
[gasp]	Inhales sharply with surprise	[gasp] Oh no! Look out behind you!
[sighs]	Exhales heavily before	[sighs] Here we go again.

	speaking	
[excited]	Increases cadence and dynamic pitch	[excited] We just launched our new flagship model!
[slow]	Reduces speaking rate by ~30%	[slow] Listen very carefully to this step.
[shouting]	Raises volume, adds projection	[shouting] Stop right there!
[sarcastically]	Inverts inflection for irony	[sarcastically] Oh, brilliant idea. What could go wrong?
[asmr]	Proximity effect with soft whisper	[asmr] Close your eyes and breathe in deeply...
[short pause]	Inserts ~500ms silent break	First step... [short pause] then the second.
[long pause]	Inserts ~1500ms silent break	The end was near. [long pause] Or so we thought.

4. The Director's Suite — Scene & Persona Context (Advanced)

To control overall vocal delivery across an entire script, expand the **Director Scene & Persona Setup** panel. Gemini 3.1 Flash constructs a holistic audio prompt using structured Markdown headers.

```
+-----+
|          DIRECTOR'S STAGE          |
+-----+
| # AUDIO PROFILE: Energetic tech presenter with infectious enthusiasm |
| ## THE SCENE: A live product keynote stadium in San Francisco        |
| ### DIRECTOR'S NOTES: Crisp consonants, vocal smile, fast pace       |
| ### TRANSCRIPT: Welcome everyone to the annual developer keynote!    |
+-----+
```

Directing Fields

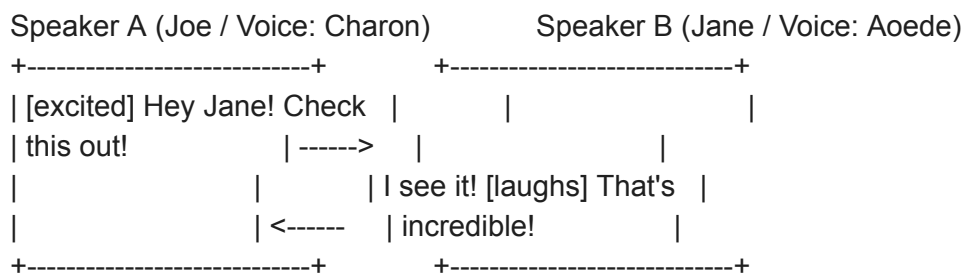
1. **Audio Profile & Character Persona:** Establishes voice identity and mood (e.g.,

"Experienced radio host with a gravelly, midnight aesthetic").

2. **The Scene / Environment:** Sets spatial and contextual acoustic cues (e.g., "A quiet library late at night", "A noisy sports stadium background").
3. **Director's Notes:** Specific technical delivery cues (e.g., "Keep pitch transitions smooth, maintain a warm vocal smile").

5. Multi-Speaker Dialogue Studio (Intermediate–Advanced)

The **Dialogue Builder** enables multi-person audio production in a single request, eliminating timing gaps between speakers.



Steps to Build Multi-Speaker Dialogues

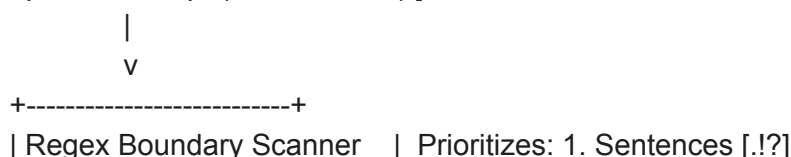
1. Switch to the **Dialogue Builder** tab.
2. Configure **Speaker 1** (Label e.g., Joe, Voice: Charon) and **Speaker 2** (Label e.g., Jane, Voice: Aoede).
3. Click **Add Speech Line** to insert conversation turns.
4. Select the speaker for each row and enter dialogue text (inline audio tags like [whispers] or [laughs] work within individual lines).
5. Click **Synthesize 3.1 Dialogue Audio**.

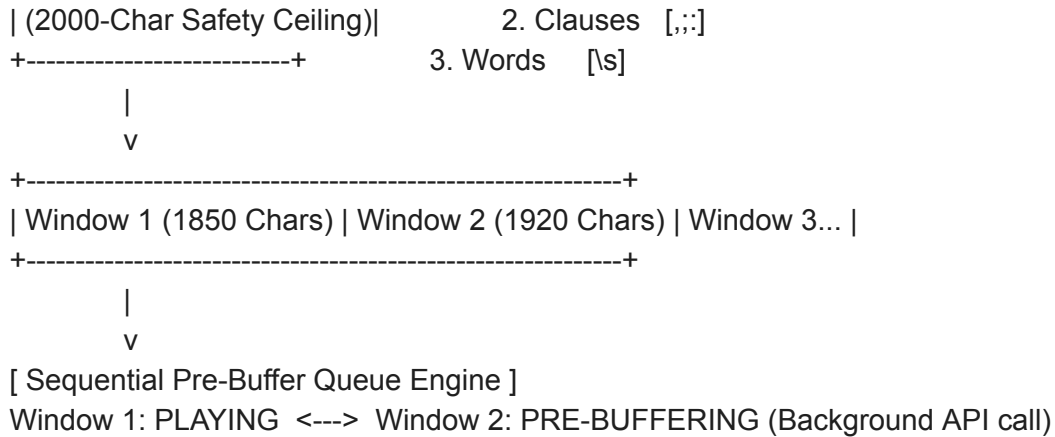
6. Technical Architecture & Under-the-Hood Mechanics (Expert)

Smart Sentence Windowing System (2,000 Chars Max / Window)

Gemini 3.1 Flash TTS limits individual request chunks for optimal latency. The app handles long documents by executing an automated **Smart Sentence-Boundary Windowing Algorithm**:

[Input Text Script (>2000 Chars)]





Windowing Algorithm Mechanics

1. If script length ≤ 2000 characters, a single window is created.
2. If script length > 2000 characters, a substring of 2,000 characters is evaluated.
3. The engine scans backward from position 2000 to locate natural boundaries:
 - o **Primary Break:** Full sentence punctuation (., !, ?) followed by whitespace.
 - o **Secondary Break:** Secondary clauses (,, ;, :).
 - o **Fallback Break:** Last space character before index 2000.
4. While **Window 1** plays in the Audio Dock, **Window 2** is fetched asynchronously in the background. When Window 1 ends, Window 2 launches seamlessly.

PCM16 to WAV Binary Container Synthesis

The Gemini API outputs raw, headerless 16-bit signed Linear PCM audio wrapped in base64 (audio/L16). Standard HTML <audio> elements cannot play raw PCM directly.

The studio implements client-side binary header synthesis using an ArrayBuffer DataView to prepend a **44-byte RIFF/WAV Header**:

0x00 [4 Bytes] 'RIFF' Chunk Identifier
 0x04 [4 Bytes] Chunk Size = 36 + PCM Data Size
 0x08 [4 Bytes] 'WAVE' Format
 0x0C [4 Bytes] 'fmt ' Subchunk Identifier
 0x10 [4 Bytes] Subchunk1 Size = 16 (for Linear PCM)
 0x14 [2 Bytes] Audio Format = 1 (Uncompressed PCM)
 0x16 [2 Bytes] Number of Channels = 1 (Mono)
 0x18 [4 Bytes] Sample Rate = 24000 Hz (parsed from mimeType)
 0x1C [4 Bytes] Byte Rate = SampleRate * NumChannels * (BitsPerSample / 8)
 0x20 [2 Bytes] Block Align = NumChannels * (BitsPerSample / 8) = 2
 0x22 [2 Bytes] Bits Per Sample = 16
 0x24 [4 Bytes] 'data' Subchunk Identifier

0x28 [4 Bytes] PCM Data Size (In Bytes)

0x2C [N Bytes] Signed Int16 Raw PCM Buffer Audio Data

Byte Rate & Alignment Formulas

$$\text{ByteRate} = \text{SampleRate} \times \text{NumChannels} \times \left(\frac{\text{BitsPerSample}}{8} \right) = 24000 \times 1 \times 2 = 48,000 \text{ Bytes/sec}$$

$$\text{BlockAlign} = \text{NumChannels} \times \left(\frac{\text{BitsPerSample}}{8} \right) = 1 \times 2 = 2 \text{ Bytes/sample}$$

This binary buffer is converted into a native Blob:

```
const wavBlob = new Blob([buffer], { type: 'audio/wav' });  
const audioUrl = URL.createObjectURL(wavBlob);
```

API Payload Schema Reference

Single Speaker Endpoint Payload (gemini-3.1-flash-tts-preview)

```
{  
  "contents": [  
    {  
      "parts": [  
        {  
          "text": "# AUDIO PROFILE: Energetic presenter\n### TRANSCRIPT:\nHello world!"  
        }  
      ]  
    }  
  ],  
  "generationConfig": {  
    "responseModalities": ["AUDIO"],  
    "speechConfig": {  
      "voiceConfig": {  
        "prebuiltVoiceConfig": {  
          "voiceName": "Kore"  
        }  
      }  
    }  
  },  
  "model": "gemini-3.1-flash-tts-preview"  
}
```

Multi-Speaker Endpoint Payload

```
{
  "contents": [
    {
      "parts": [
        {
          "text": "Joe: [excited] Hey Jane!\nJane: [whispers] Keep it down!"
        }
      ]
    }
  ],
  "generationConfig": {
    "responseModalities": ["AUDIO"],
    "speechConfig": {
      "multiSpeakerVoiceConfig": {
        "speakerVoiceConfigs": [
          {
            "speaker": "Joe",
            "voiceConfig": { "prebuiltVoiceConfig": { "voiceName": "Charon" } }
          },
          {
            "speaker": "Jane",
            "voiceConfig": { "prebuiltVoiceConfig": { "voiceName": "Aoede" } }
          }
        ]
      }
    }
  },
  "model": "gemini-3.1-flash-tts-preview"
}
```

7. Troubleshooting & Operator FAQ

Common Issues & Resolutions

1. Error 429 (Rate Limit Exceeded):

- *Cause:* Sending requests too quickly.
- *Resolution:* The application built-in fetchWithRetry utility automatically applies exponential backoff logic (**1s, 2s, 4s**). Wait a few seconds before retrieving next

batch.

2. **Audio Not Auto-Playing in Browser:**

- *Cause:* Browser Web Audio API policies block un-muted audio before user interaction.
- *Resolution:* Click anywhere on the studio page before triggering your first generation to initialize the global AudioContext.

3. **Inconsistent Pitch Between Smart Windows:**

- *Cause:* When long text is split across multiple windows, natural inflections can reset at window boundaries.
- *Resolution:* Populate the **Director Profile & Scene** fields. Setting global environmental context ensures consistent tone across all sentence windows.